CISS350: Data Structures and Advanced Algorithms
Quiz q10702

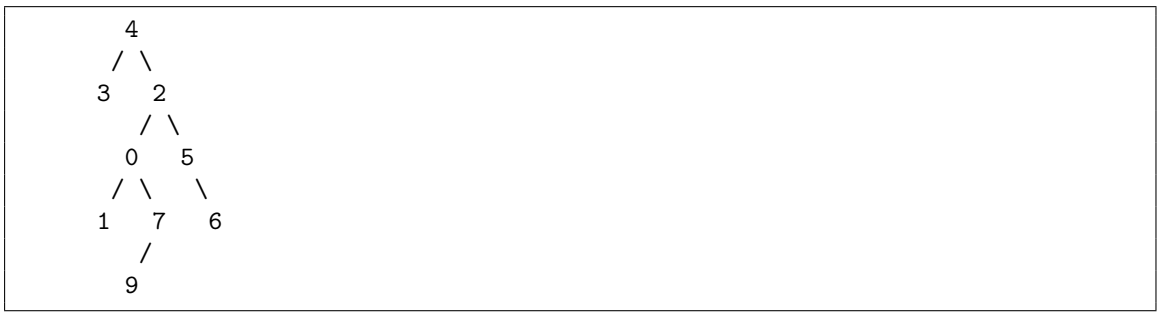
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You are given the following binary tree node class:

```
class BTNode
{
public:
    int key_;
    BTNode * parent_, * left_, * right_;
};
```

The following questions uses this class. Make sure you test your code.

Q1. In the following tree



assume that the tree contains unique values. Suppose p points to the node with value 4. Then the tree that p points to (i.e., the root of the tree that p points to) has height 4. If p points to the node with value 2. Then the tree that p points to has height 3. Complete the following function that returns the height as described above.
Answer:

```
int height(BTNode * p)
{
  if(p == NULL)
      return 0;

  int leftH = height(p->left_);
  int rightH = height(p->right_);

  return 1 + (leftH > rightH ? leftH : rightH);
}
```

Q2. In the following tree



assume that the tree contains unique values.

- Suppose `p` points to the node with value 4 and `q` points to the node with value 5. Then we say that 5 is at a depth of 2 below 4 and when I call `depth(p, q)`, I get 2.
- If `p` points to the node with value 2 and `q` points to the node with value 9. Then we say that 9 is at a depth of 3 below 2 and when I call `depth(p, q)`, I get 3.
- If `p` points to the node with value 2 then `depth(p, p)` returns 0.

The above describes the cases when `p` and `q` points to nodes and either `p` and `q` are the same or `*q` is a descendent of `*p`. In all other cases, the function should return -1. Complete the following function that returns the depth as described above.
Answer:

```
int depth(BTNode * p, BTNode * q)
{
  if(p == NULL && q == NULL
      return -1;
  if(p == q)
      return 0;

  BTNode * curr = q;
  int d = 0;

  while(curr != NULL)
  {
      if(curr == p)
         return d;

      curr = curr->parent_;
      ++d;
  }

  return -1;
}
```

Q3. In the following tree



assume that the tree contains unique values.

- Suppose `p` points to the node with value 4, the depth of 0 (below 4) is said to be 2. The depth of 6 (below 4) is 3.

Check the notes. Complete the following function that returns the depth of the node with `key_` value of `target` below the node that `p` points to. If `target` cannot be found, `-1` is returned.

Answer:

```
int depth(BTNode * p, int target)
{
  if(p == NULL)
    return -1;

   std::queue<BTNode*> nodeq;
   std::queue<int> depthq;

   nodeq.push(p);
   depthq.push(0);

   while(!nodeQ.empty())
   {
     BTNode * curr = nodeq.front();
     nodeq.pop();

     int d = depthq.front();
     depthq.pop();

     if (curr->key_ == target)
       return d;

     if (curr->left_ != NULL)
     {
       nodeQ.push(curr->left_);
       depthQ.push(d + 1);
     }

     if (curr->right_ != NULL)
     {
       nodeQ.push(curr->right_);
       depthQ.push(d + 1);
```

```
    }
  }

  return -1;
}
```

(Hint: Do a BF traversal. Instead of putting node pointers into the queue, you want to put the node pointer together with its depth into the queue. Note that even if the tree has repeat `key_` values, if you do a BF traversal, the depth computed is the *shallowest* depth of all the nodes with the `key_` value of `target`.)

Q4. Complete the function below that returns true if `p` and `q` point to roots of binary trees which are the same. In this case, same means the tree has the same structure, i.e., the root nodes have the same `key_` value, while left child nodes have the same value, then right child node have the same `key_` value, etc. And if a node in the tree `p` points to does not have a left child, then the corresponding node in the tree `q` points to does not have a left child; likewise for right child. The following trees are the same

```
      4            4
     / \          / \
    3   2        3   2
       / \          / \
      0   5        0   5
```

The following trees are different:

```
      4            4
     / \          / \
    3   2        3   2
       / \          / \
      0   5        5   0
```

The following trees are different:

```
      4            4
     / \          / \
    3   2        3   2
       / \            \
      0   5            5
```

Don't forget to check the parent pointer.

Answer:

```
bool is_same_tree(BTNode * p, BTNode * q)
{
  if(p == NULL && q == NULL) return true;
  if(p == NULL || q == NULL) return false;
  if(p->key_ != q->key_) return false;

  if ((p->parent_ == NULL && q->parent_ != NULL) ||
```

```
     (p->parent_ != NULL && q->parent_ == NULL))
    return false;

  if (p->parent_ != NULL && q->parent_ != NULL &&
      p->parent_->key_ != q->parent_->key_)
    return false;

  return is_same_tree(p->left_, q->left_) &&
         is_same_tree(p->right_, q->right_);
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.